Voltage Drop Chart

How far each copper wire size can run before it loses 3 percent of the voltage, at common currents and voltages.

For a 3 percent drop, the longest one-way run is 0.03 times the voltage times the wire area, divided by the current and a constant. At 120 V, 12 AWG copper (3.31 mm²) carrying 20 A reaches the 3 percent mark at 13 m. The same wire on a 240 V circuit goes twice as far, because the allowed drop in volts is twice as large.

Maximum one-way length for 3% drop: 120 V single-phase

Wire size	5 A	10 A	15 A	20 A	30 A	40 A	50 A
14 AWG	34	17	11	—	—	—	—
12 AWG	55	27	18	13	—	—	—
10 AWG	88	44	29	22	14	—	—
8 AWG	140	70	46	35	23	17	14
6 AWG	223	111	74	55	37	27	22
4 AWG	355	177	118	88	59	44	35
3 AWG	447	223	149	111	74	55	44
2 AWG	564	282	188	141	94	70	56
1 AWG	711	355	237	177	118	88	71
1/0 AWG	898	449	299	224	149	112	89
2/0 AWG	1,132	566	377	283	188	141	113
3/0 AWG	1,427	713	475	356	237	178	142
4/0 AWG	1,799	899	599	449	299	224	179

Copper conductors, in metres, rounded down. A dash means that current is more than the conductor may carry, so no length is given.

Maximum one-way length for 3% drop: 240 V single-phase

Wire size	15 A	20 A	30 A	40 A	50 A	60 A	80 A	100 A
14 AWG	23	—	—	—	—	—	—	—
12 AWG	37	27	—	—	—	—	—	—
10 AWG	58	44	29	—	—	—	—	—
8 AWG	93	70	46	35	28	—	—	—
6 AWG	148	111	74	55	44	37	—	—
4 AWG	236	177	118	88	71	59	44	—
3 AWG	298	223	149	111	89	74	55	44
2 AWG	376	282	188	141	112	94	70	56
1 AWG	474	355	237	177	142	118	88	71
1/0 AWG	598	449	299	224	179	149	112	89
2/0 AWG	754	566	377	283	226	188	141	113
3/0 AWG	951	713	475	356	285	237	178	142
4/0 AWG	1,199	899	599	449	359	299	224	179

Copper conductors, in metres, rounded down. A dash means that current is more than the conductor may carry, so no length is given.

Maximum one-way length for 3% drop: 480 V three-phase

Wire size	20 A	30 A	50 A	60 A	80 A	100 A	150 A	200 A
14 AWG	—	—	—	—	—	—	—	—
12 AWG	64	—	—	—	—	—	—	—
10 AWG	101	67	—	—	—	—	—	—
8 AWG	162	108	64	—	—	—	—	—
6 AWG	257	171	103	85	—	—	—	—
4 AWG	409	273	163	136	102	—	—	—
3 AWG	516	344	206	172	129	103	—	—
2 AWG	651	434	260	217	162	130	—	—
1 AWG	821	547	328	273	205	164	—	—
1/0 AWG	1,037	691	414	345	259	207	138	—
2/0 AWG	1,307	871	522	435	326	261	174	—
3/0 AWG	1,648	1,098	659	549	412	329	219	164
4/0 AWG	2,078	1,385	831	692	519	415	277	207

Copper conductors, in metres, rounded down. A dash means that current is more than the conductor may carry, so no length is given.

How the values are calculated

$$L = (0.03 \times V \times A) / (2 \times 0.0214 \times I) \quad \text{[single-phase]}$$

$$L = (0.03 \times V \times A) / (1.732 \times 0.0214 \times I) \quad \text{[three-phase]}$$

L in metres, A in mm², copper at about 75°C

Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

The K-factor method uses resistance only. It ignores reactance, which matters on large conductors and long runs, so the lengths for the biggest sizes are optimistic. It assumes copper at about 75°C and a load at the far end. The 3 percent figure is a recommendation in the NEC informational notes for a branch circuit or a feeder, not a requirement, though some equipment and some local codes do require it.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/voltage-drop-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.